TECHNICAL ARCHITECTURE & SPECIFICATION DOCUMENT

Career Compass

AI-Powered Career Guidance, Resume Analysis & Technical Interview Platform

DOCUMENT VERSION

Version 1.1 (Production Aligned)

STATUS

Active Production Codebase & Roadmap

CORE TECH STACK

FastAPI, React 18, Ollama, Celery, MySQL & Mongo

DOCUMENTATION TYPE

SRS, Architecture & Developer Reference

This document serves as the authoritative technical specification and architectural blueprint for the Career Compass platform. It accurately details all implemented system components, data schemas, API contracts, and the strategic expansion roadmap.

1. Executive Summary & System Purpose

Career Compass is a full-stack, AI-driven career guidance and resume intelligence platform. Unlike traditional keyword-matching career tools, Career Compass integrates local Generative AI models (via Ollama running `llama3.2`, with cloud OpenAI fallback) to deeply understand candidate backgrounds, extract nuanced skills and achievements, benchmark profiles against industry standards, and generate structured 3-month skill acquisition roadmaps.

This document provides a comprehensive technical reference for the platform. It documents both the **currently implemented production features (Version 1.0)** and the **planned feature expansion roadmap (Version 2.0+)**, ensuring complete transparency for developers, contributors, and evaluators.

2. User Roles & System Permissions

- **Guest (Unauthenticated User):** Can browse landing information, view public shareable analysis links (via token), create an account, authenticate, and request password resets.
- **Standard User (Job Seeker / Professional):** Can upload PDF resumes, manage personal profile details, manually input experience, perform skill gap analyses, take technical skill assessments, complete mock interviews, view past history, export bilingual PDF reports, and toggle anonymous benchmarking.
- **System Administrator (Future Scope):** Dedicated administrative portal to manage users, monitor analytics, audit AI inference latency, and tune recruitment prompt templates.

3. System Architecture & Technology Stack

3.1 Backend Architecture

- **Web Framework:** FastAPI (Python 3.11+) utilizing asynchronous request handling, Pydantic validation schemas, and CORS security middleware.
- **Relational Database (SQLAlchemy ORM):** Hybrid database architecture utilizing MySQL in production with automated fallback to SQLite for local development and zero-dependency automated testing. Manages structured catalogues: Jobs, Skills, Skill Weights, Assessment Questions, Question Options, and Courses.
- **Document Database (MongoDB):** Used for flexible semi-structured data including raw CV extractions, user assessment responses, recommendation results, feedback loops, and interview dialogues. Mongomock is utilized for fast in-memory testing.
- **Task Worker & Message Queue:** Celery worker backed by Redis for asynchronous execution and scheduled cron tasks (such as automated weekly career roadmap progress emails via SMTP).
- **AI & LLM Pipeline:** Local LLM execution via Ollama (`llama3.2`) ensuring low latency and data privacy, with dynamic fallback to OpenAI (`gpt-3.5-turbo`) when an API key is provided.
- **Document Processing:** `pdfplumber` for parsing English and Arabic resume text; `ReportLab` with bundled `Amiri` fonts for rendering RTL-compliant downloadable PDF career reports.

- **Security:** Passwords hashed using `bcrypt`; stateless session management enforced with JWT access tokens.

3.2 Frontend Architecture

- **UI Framework & Build:** React 18 powered by Vite for rapid HMR and optimized production builds.
- **Styling & Design System:** Tailwind CSS for a responsive, modern developer experience.
- **Routing & Access Control:** React Router v6 configured with `ProtectedRoute` guards for authenticated views.
- **State Management:** Modular React Contexts (`AuthContext`, `AnalysisContext`, `ToastContext`).
- **Visualizations:** Recharts for rendering skill match radar charts, progress circles, and peer benchmark bars.
- **Networking:** Axios client configured with centralized base URL, authorization headers, and unified error handling.

4. Database Architecture & Schemas

4.1 Relational Tables (SQLAlchemy)

The relational database manages system knowledge bases, job roles, skills, and account authentication:

- **users:** `id` (Integer, PK), `full_name` (String), `email` (String, Unique), `hashed_password` (String), `benchmark_opt_in` (Boolean), `notifications_opt_in` (Boolean), `created_at` (DateTime), `updated_at` (DateTime).
- **jobs:** `id` (Integer, PK), `title` (String), `description` (Text), `industry` (String), `level` (Enum: junior, mid, senior), `created_at` (DateTime).
- **skills:** `id` (Integer, PK), `name` (String, Unique), `category` (String), `description` (Text).
- **job_skills:** `id` (Integer, PK), `job_id` (FK → jobs.id), `skill_id` (FK → skills.id), `importance` (Enum: essential, important, bonus), `weight` (Float).
- **questions:** `id` (Integer, PK), `job_id` (FK → jobs.id), `skill_id` (FK → skills.id), `text` (Text), `type` (Enum: single_choice, multiple_choice, scale), `difficulty` (Integer).
- **question_options:** `id` (Integer, PK), `question_id` (FK → questions.id), `text` (String), `score_value` (Float), `is_correct` (Boolean).
- **courses:** `id` (Integer, PK), `skill_id` (FK → skills.id), `title` (String), `provider` (String), `url` (String), `duration_hours` (Integer), `difficulty` (String).

4.2 Document Collections (MongoDB)

MongoDB manages user-generated artifacts and dynamic analysis payloads:

- **uploads:** `file_id` (UUID), `user_id` (Integer), `filename`, `path`, `uploaded_at`.

- **extractions:** `extraction_id` (UUID), `user_id`, `file_id`, `skills` (Array), `experience` (Array), `education` (Array), `projects` (Array), `confirmed` (Boolean).
- **assessments:** `assessment_id` (UUID), `user_id`, `answers` (Array of question_id, selected_option, score), `submitted_at`.
- **analysis_results:** `analysis_id` (UUID), `user_id`, `extraction_id`, `assessment_id`, `job_match` (match score, top job, alternatives), `skill_gaps`, `roadmap` (3-month schedule), `career_score`, `checklist_progress`, `feedback`, `shared` (token, is_active).
- **interviews:** `session_id` (UUID), `user_id`, `analysis_id`, `job_title`, `messages` (Array of role/content), `created_at`, `status`.

5. API Endpoints Specification (Current Codebase)

CATEGORY	METHOD	ENDPOINT	FUNCTIONALITY	AUTH
Auth	POST	`/register`	User signup (name, email, password)	No
Auth	POST	`/login`	User authentication and JWT token issuance	No
Auth	POST	`/forgot-password`	Issue password reset token via email	No
Auth	POST	`/reset-password`	Apply new password with verified token	No
Profile	GET	`/profile`	Fetch user profile, career stats, and target role	Yes
Profile	PUT	`/profile`	Update full name and target job role	Yes
Profile	GET	`/profile/badges`	Retrieve gamified achievement badges	Yes
Profile	DELETE	`/profile`	Permanently delete account and all history	Yes
Settings	PUT	`/settings/password`	Change current account password	Yes
Settings	PUT	`/settings/benchmark-opt-in`	Toggle anonymous peer comparison sharing	Yes
Settings	PUT	`/settings/notifications`	Toggle weekly Celery progress update emails	Yes
CV	POST	`/upload-cv`	Upload and validate resume PDF (≤ 5MB)	Yes
CV	POST	`/extract`	Extract structured data via Ollama LLM	Yes
CV	POST	`/extract/manual`	Direct manual input of skills and experience	Yes
CV	POST	`/extract/confirm`	Confirm and adjust extracted profile data	Yes

CATEGORY	METHOD	ENDPOINT	FUNCTIONALITY	AUTH
CV	GET	`/extract/{id}/ats-score`	ATS compatibility scoring and audit breakdown	Yes
CV	POST	`/extract/{id}/cover-letter`	AI Cover Letter generation tailored to job post	Yes
Assessment	GET	`/assessment`	Fetch role-specific technical questions	Yes
Assessment	POST	`/assessment`	Submit answers and record technical score	Yes
Analysis	POST	`/analyze`	Execute recommendation engine and roadmap	Yes
History	GET	`/history`	List historical analyses sorted by date	Yes
History	GET	`/history/{id}`	Fetch single analysis full report	Yes
History	GET	`/history/compare`	Compare two analyses side-by-side	Yes
History	GET	`/history/{id}/export/pdf`	Export branded Arabic/English PDF report	Yes
History	PATCH	`/history/{id}/roadmap`	Update roadmap checklist milestone progress	Yes
History	POST	`/history/{id}/feedback`	Submit recommendation rating and notes	Yes
History	POST	`/history/{id}/share`	Create public shareable link	Yes
History	DELETE	`/history/{id}/share`	Revoke public shareable link	Yes
Shared	GET	`/shared/{token}`	Public unauthenticated report view	No
Benchmark	GET	`/benchmark`	Anonymous cohort comparison metrics	Yes
Interview	POST	`/interview/start`	Start AI technical mock interview session	Yes
Interview	POST	`/interview/chat`	Interactive chat with the AI interviewer	Yes

6. Detailed Implemented Features (Version 1.0)

6.1 AI Resume Extraction & Manual Fallback

The platform validates uploaded files (MIME type `application/pdf`, maximum size 5MB) and uses `pdfplumber` to extract text from multi-column layouts in both Arabic and English. Extracted text is fed to Ollama (`llama3.2`) to yield structured JSON objects mapping skills, work history, education, and projects. For users without a CV, a comprehensive manual entry interface allows direct profile specification.

6.2 Dynamic Recommendation Engine & 3-Month Roadmap

The recommendation algorithm matches candidate skills against database job profiles, applying categorized weights (essential: 1.0, important: 0.6, bonus: 0.3). It evaluates technical quiz performance to calibrate proficiency, generating:

- **Overall Career Match Score:** Percentage readiness for target and alternative roles.
- **Skill Gap Identification:** Clear breakdown of acquired competencies vs. missing requirements.
- **Structured 3-Month Roadmap:** Monthly progressive milestones pairing missing skills with recommended curated online courses (Coursera, Udemy, edX) and practical project exercises.
- **Interactive Milestone Tracking:** Users can check off acquired skills directly on their roadmap, updating database records in real time.

6.3 Interactive Mock Interview Simulator

Simulates a real technical interview where the AI adopts the persona of a hiring manager for the candidate's target role. Questions are dynamically tailored to the candidate's specific background and identified skill deficiencies.

6.4 ATS Audit & AI Cover Letter Generator

Evaluates resumes against automated applicant tracking system (ATS) criteria, highlighting formatting risks, section hierarchy issues, and missing industry keywords. Provides a one-click AI cover letter generator tailored to target job descriptions.

6.5 Gamification, Peer Benchmark & Export

Includes user achievement badges (e.g., Profile Pioneer, Roadmap Master), anonymous peer benchmarking percentiles, and full PDF export supporting bidirectional Arabic text rendering using bundled Amiri typography.

7. Strategic Expansion Roadmap (Version 2.0+ Future Scope)

The following high-impact features represent the planned expansion trajectory for future versions:

7.1 Phase 2: Advanced AI & User Experience (Next Horizon)

- **Celery Asynchronous PDF Worker:** Migrating CV parsing into a decoupled background task worker (`parse_cv_task`) returning immediate task IDs, paired with client polling (`/cv/status/{task_id}`) to handle heavy upload traffic seamlessly.
- **Structured 10-Point Interview Scoring:** Dedicated evaluation endpoint (`/interview/evaluate`) that grades submitted answers on a 10-point scale, flags specific grammatical and technical errors, and generates model answer suggestions.

- **Voice & Audio Mock Interviews:** Integration with OpenAI Whisper to allow candidates to speak answers aloud, providing AI feedback on speech clarity, pace, and vocal confidence.
- **LinkedIn Profile Import:** Parser specialized in reading standard LinkedIn PDF profile exports directly into the Career Compass profile schema.
- **Full ATS CV Builder & Exporter:** In-browser resume builder that formats and exports ATS-compliant resume PDFs directly from refined profile data.
- **Live Market Job Matching:** Direct integration with live job boards (via APIs or scrapers) to display open vacancies matching the user's updated skill set.

7.2 Phase 3: B2B Expansion & Commercialization

- **B2B HR & Recruiter Portal:** A dedicated corporate dashboard for employers to post job specifications and automatically discover, filter, and rank matching candidate CVs.
- **Monetization & Payment Gateways:** Integration with Stripe and PayPal to provide tiered subscriptions (Freemium vs. Unlimited AI Pro).
- **Chrome Browser Extension:** Browser extension that reads active job listings on LinkedIn, compares them against the user's CV, and displays immediate compatibility scores.
- **Mobile Application:** Cross-platform mobile app (React Native / Flutter) for practicing mock interviews and checking roadmap milestones on mobile devices.
- **Domain Fine-Tuned LLM:** Fine-tuning open-source LLMs specifically on regional recruitment standards and technical hiring taxonomies.